

Equation Provenance Annex

AI-Powered Unified Mobility and Delivery System Using Dynamic Feasibility Analysis

Supporting note — not part of the submitted manuscript

How to read this. **Original** means the relation is defined in this paper. **Standard form, original instantiation** means the mathematical shape is conventional and the contribution is what it is applied to. **Classical** means the result has a named origin in the literature and is cited.

Eq.	What it states	Origin	Basis
(1)	Capability-coverage index CCI	Original (evaluation device)	Defined in this paper to score Table I; not claimed as a research contribution
(2)	Compatibility score $CS(i, j)$	Standard form, original instantiation	Weighted linear scalarisation, as in multi-criteria decision analysis; the five criteria and their use for tri-modal dispatch are this paper's
(3)	Admission rule $CS \geq \tau$	Original	Threshold rule on (2)
(4)	Time-window overlap $T(i, j)$	Standard form, original instantiation	Triangular (tent) similarity kernel; the service-type tolerance W_{type} is contributed here
(5)	Capacity indicator $V(i, j)$	Standard	Capacity feasibility indicator conventional in capacitated vehicle routing
(6)	Priority compatibility $Q(i, j)$	Standard form	Normalised-difference similarity measure
(7)	Route objective $\min_{\pi}[\cdot]$	Standard form	Weighted-sum scalarisation of a pickup-and-delivery problem with time windows
(8)	Assignment cost $Cost(d, \rho)$	Standard form	Weighted linear cost over distance, ETA and capacity mismatch
(9)	Greedy selection d^*	Standard	arg min over available drivers
(10)	Boundedness $CS \in [0, 1]$	Original (proved)	Convexity of the combination in (2)
(11)	Lipschitz bound on T	Original application	Lipschitz continuity (classical analysis) applied to (4); the service-type asymmetry is this paper's result
(12)	Fail-closed bound, Prop. 1	Original (proved)	Proved in this paper from the threshold rule and solver feasibility
(13)	Complexity $O(\mathcal{R} k)$	Original statement	Bounded k -neighbourhood argument applied to (2)
(14)	Linear assignment problem	Classical — cited	Kuhn, <i>Naval Res. Logist. Quart.</i> , 1955 (ref. [12]); Hungarian method
(15)	Softmax α_k	Standard	Softmax parameterisation onto the probability simplex
(16)	Binary cross-entropy loss	Standard	Standard classification loss
(17)	$\partial CS / \partial \alpha_k = F_k$	Trivial	Derivative of the linear form (2)
(18)	Softmax Jacobian	Standard	$\alpha_k(\delta_{km} - \alpha_m)$, textbook result
(19)	Gradient update on θ	Standard	Stochastic gradient descent
(20)	Threshold utility $\mathbb{E}[U(\tau)]$	Standard form	Expected utility of a decision threshold, as in ROC analysis

Summary. Five relations are original to the paper: (1), (3), (10), (12) and (13), together with the specific instantiation of (2) and the service-type asymmetry established through (11). One is classical and cited: (14). The remainder are conventional constructions from multi-criteria decision analysis, vehicle routing, and machine learning, applied to this problem. The paper's contribution is their composition into a feasibility-first decision layer spanning passenger, food and parcel requests, and the properties proved in Section III-D. Section III-F of the manuscript states this in the same terms.

If asked in review or viva: the honest position is that no new mathematics is claimed. What is claimed is the feasibility-first ordering, the five-criterion score for three service types with per-type timing tolerance, and the three proved properties — boundedness, the Lipschitz asymmetry, and the fail-closed bound.